

Remote Sensing Analysis

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Summary

This chapter provides satellite-based evidence on the effects of CFM on fire, deforestation and above-ground biomass across the surveyed community forests and their surrounding buffer zones. Analyses use fire detection data (MODIS), tree cover loss data (Hansen Global Forest Change) and radar-based biomass estimates, employing before-after mixed models with a 1 km buffer zone as a spatial control to assess leakage.

Key conclusions

Across all three fire metrics — detection density, burned area fraction and fire radiative power — there were statistically significant reductions within CF cores following establishment. These results constitute robust, multi-indicator satellite confirmation that CFM has led to meaningful improvements in fire management across the landscape. Crucially, none of the fire models showed a significant Before–After $\times$ Zone interaction, indicating that there was no measurable displacement of burning from CF cores into the surrounding buffer zones. The absence of leakage is an important positive finding: it suggests that community fire management extends beyond the formal CF boundary, consistent with the community perceptions documented in the SFM chapter.

According to the Hansen Global Forest Canopy dataset, tree cover loss rates within CF cores did not decrease significantly following establishment. This result should be interpreted cautiously. Baseline deforestation rates within CF areas were already very low (median approximately 0.14% per year before establishment), leaving limited statistical scope for further measurable decline; nevertheless, the directional trend is in the right direction. Tree cover data are also known to be unreliable at small spatial scales in miombo woodlands, where canopy cover is naturally sparse and can vary seasonally. Communities may be perceiving real reductions in forest degradation through indirect indicators — reduced charcoal production, fewer incursions, recovering understorey — that are not captured by the canopy loss metric. The significant increase in tree cover loss observed in the 1 km buffer zones after CF establishment is a concern, but should also be interpreted with caution given these measurement uncertainties. Continued monitoring over a longer period is necessary before strong conclusions can be drawn on deforestation

trends.

Above-ground biomass increased significantly in both CF cores and buffer zones following CFMG establishment. For dry seasonal forests such as the miombo woodlands of Zambia, biomass is likely a more sensitive and ecologically meaningful indicator of forest condition than canopy cover loss, as it can detect incremental changes in tree recruitment and growth that gross deforestation metrics miss. The observed biomass gains are consistent with — and likely substantially attributable to — the documented reductions in fire disturbance. Fire both kills trees directly and suppresses woody recruitment; even modest reductions in annual burning can produce measurable biomass accumulation over a period of three to seven years.

The Governance Index was not a significant predictor of any remote sensing outcome, in contrast to the SFM chapter where governance quality was a strong predictor of community perceptions. This may reflect that basic fire management is among the first activities initiated by newly established CFMGs, and that even groups with weaker governance have implemented fire patrols — the primary mechanism linking CFM to the observed fire reductions.

Taken together, the remote sensing evidence presents a broadly positive picture of CFM effectiveness. CFM has demonstrably reduced fire disturbance and produced measurable gains in above-ground biomass across both CF cores and immediate buffer zones, with no evidence of displacement of pressure into surrounding areas. The deforestation signal remains ambiguous and warrants continued monitoring. The most important implication is that the fire management benefits of CFM are real and landscape-scale, but the economic drivers of deforestation — charcoal production, agricultural clearance — remain largely unaddressed by current CFM practice.

Introduction

The preceding chapter examined the effects of community forestry on sustainable forest management based on community perceptions collected through focus group discussions and individual interviews. In this chapter we provide an independent assessment of CFM impacts using satellite-derived metrics. Remote sensing offers the advantage of objective, spatially consistent measurements that are free from reporting biases, and captures processes that may be difficult for community members to perceive and quantify.

We analysed three categories of remote sensing data. Fire dynamics were assessed using MODIS active fire detection data (Giglio, Schroeder, and Justice 2016), from which we derived three complementary metrics: fire detection density (detections per km² per year), burned area as a fraction of total zone area, and total fire radiative power (FRP) per km² — a measure of fire energy release that reflects combustion intensity and therefore the volume of biomass consumed. Deforestation was quantified using Hansen Global Forest Change (GFC) 2024 tree cover loss data (Hansen et al. 2013), expressed as the annual percentage of zone

Table 1: Linear mixed model results for log-transformed annual fire count before and after CF establishment.

Model results for fire detection density (log-transformed) — before vs after establishment

Linear mixed model with random intercept for CFMG. Response = $\log(\text{detections per km}^2 + 1)$. Fixed effects: period (ref = Before) and governance index. CF variance = 0.145; residual variance = 0.191. Marginal $R^2 = 0.055$; conditional $R^2 = 0.463$. $n = 817$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before, Core)	1.098	0.161	0.0000
After	-0.121	0.046	0.0083
Buffer	-0.303	0.040	0.0000
Governance index	-0.007	0.030	0.8290
After $\times$ Buffer	0.095	0.062	0.1278

area experiencing tree cover loss. Above-ground biomass (AGB) was assessed using annual remotely sensed biomass estimates based on radar backscatter covering 2018–2024 (Harrison et al. in prep).

For each metric, we compared values before and after CF establishment, using the formal establishment date of each CFMG. Fire and deforestation data covered the period 2014–2024. All analyses used linear mixed models (or, for the proportion-bounded burned area response, a Gamma generalised linear mixed model with a log link), with a random intercept for CFMG to account for the non-independence of repeated annual observations within the same CF. We included the CFMG governance index as a fixed covariate to test whether governance quality moderated outcomes. To assess leakage — the displacement of forest-damaging activities into the areas surrounding CFs — we included a 1 km buffer zone around each CF as a spatial control and tested whether the Before–After change differed between core and buffer zones (the period $\times$ zone interaction). A non-significant interaction would indicate that changes in the buffer mirrored those in the CF.

Fire incidence

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On average, fire incidence was significantly reduced following the establishment of a CF and this effect was similar inside the CF and within its buffer zone (i.e. no significant period $\times$ location interaction). Interestingly, on average fire incidence was lower in the buffer zone compared to the CF. The Governance Index did not have an significant effect on these outcomes.

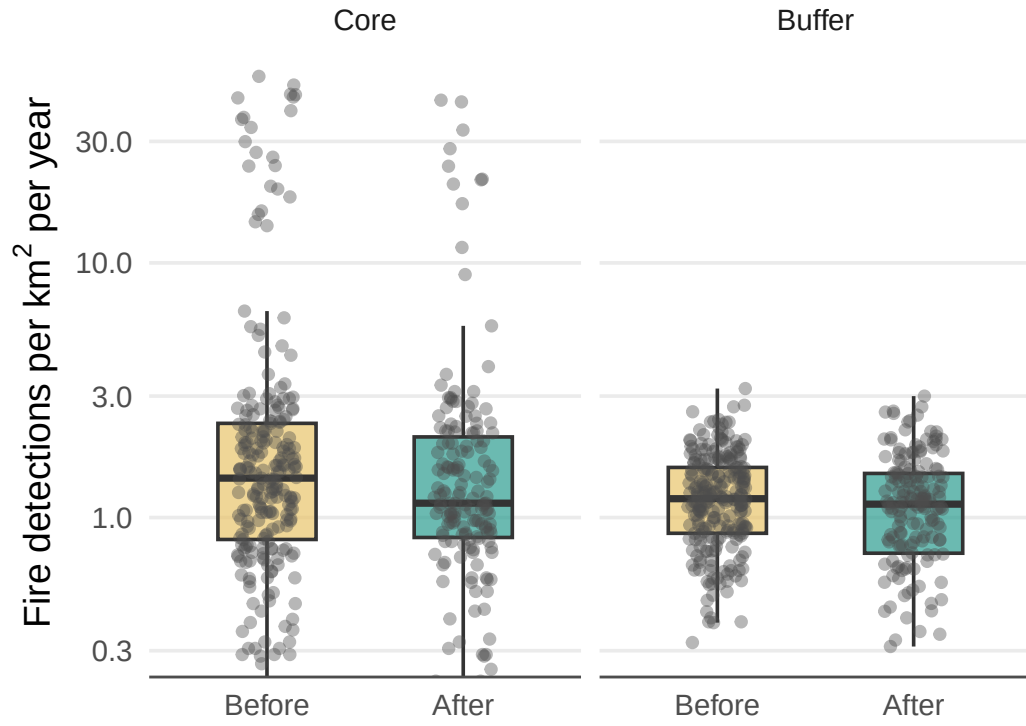


Figure 1: Annual fire detection density (detections per km^2) per CF before and after establishment, for CF cores and 1 km buffer zones. Each point represents one CF-year. Boxes show median and interquartile range.

Burned area

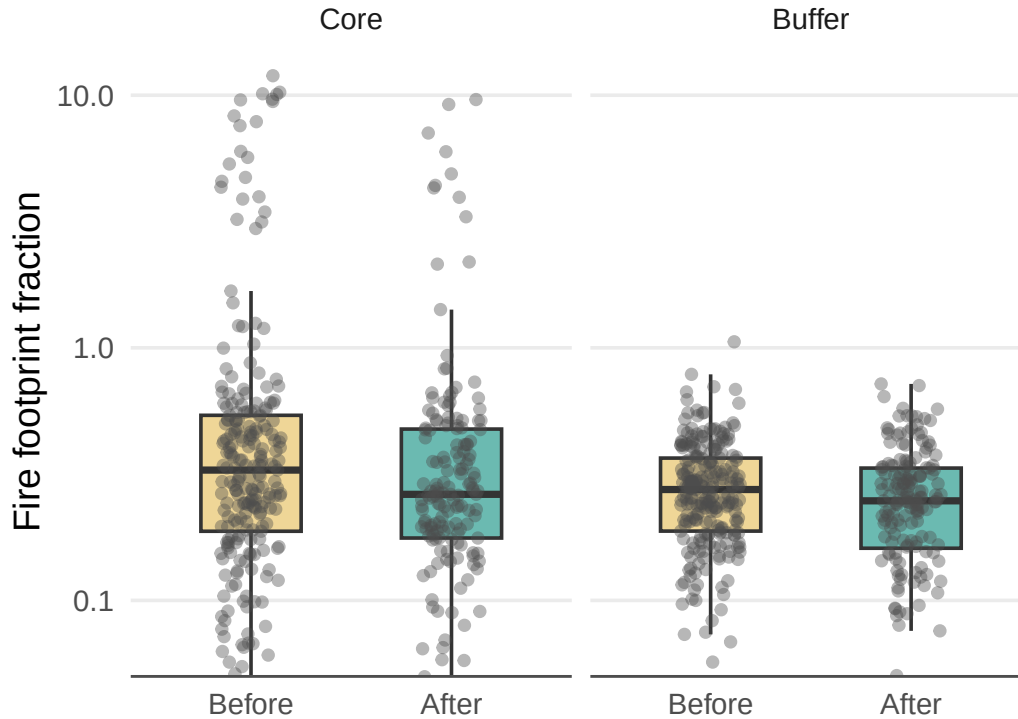


Figure 2: Annual fire footprint as a fraction of area before and after CF establishment, for cores and 1 km buffers. Fire footprint estimated as the sum of scan $\times$ track for all detections within each zone-year, divided by zone area. Each point represents one CFMG-year. Boxes show median and interquartile range.

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The results for burned area exactly mirrored those obtained for fire incidence. The burned area declined following establishment of community forestry in both CFs and buffer zones. There was no significant effect of Governance Index on the burned area.

Fire radiative power

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Again, the results for fire radiative power mirrored those obtained for fire incidence.

Table 2: Gamma GLMM results for annual fire footprint fraction before and after CF establishment.

Model results for fire footprint fraction — Core vs Buffer, before vs after establishment

Gamma GLMM with log link and random intercept for CFMG. Fixed effects: period (ref = Before), zone (ref = Core), period $\times$ zone interaction (After $\times$ Buffer = leakage test), and governance index. CFMG variance = 0.582. Marginal $R^2 = 0.032$; conditional $R^2 = 0.68$. $n = 817$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before, Core)	-0.774	0.307	0.0118
After	-0.181	0.068	0.0073
Buffer	-0.318	0.062	0.0000
Governance index	-0.026	0.058	0.6603
After $\times$ Buffer	0.086	0.091	0.3481

Table 3: Linear mixed model results for log-transformed annual total FRP per km² before and after CF establishment.

Model results for total FRP per km² (log-transformed) — Core vs Buffer, before vs after establishment

Linear mixed model with random intercept for CFMG. Response = $\log(\text{total FRP} / \text{zone area} + 1)$. Fixed effects: period (ref = Before), zone (ref = Core), period $\times$ zone interaction (After $\times$ Buffer = leakage test), and governance index. CFMG variance = 0.381; residual variance = 0.549. Marginal $R^2 = 0.029$; conditional $R^2 = 0.426$. $n = 817$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before, Core)	2.748	0.261	0.0000
After	-0.219	0.078	0.0050
Buffer	-0.298	0.068	0.0000
Governance index	-0.028	0.049	0.5738
After $\times$ Buffer	0.110	0.105	0.2991

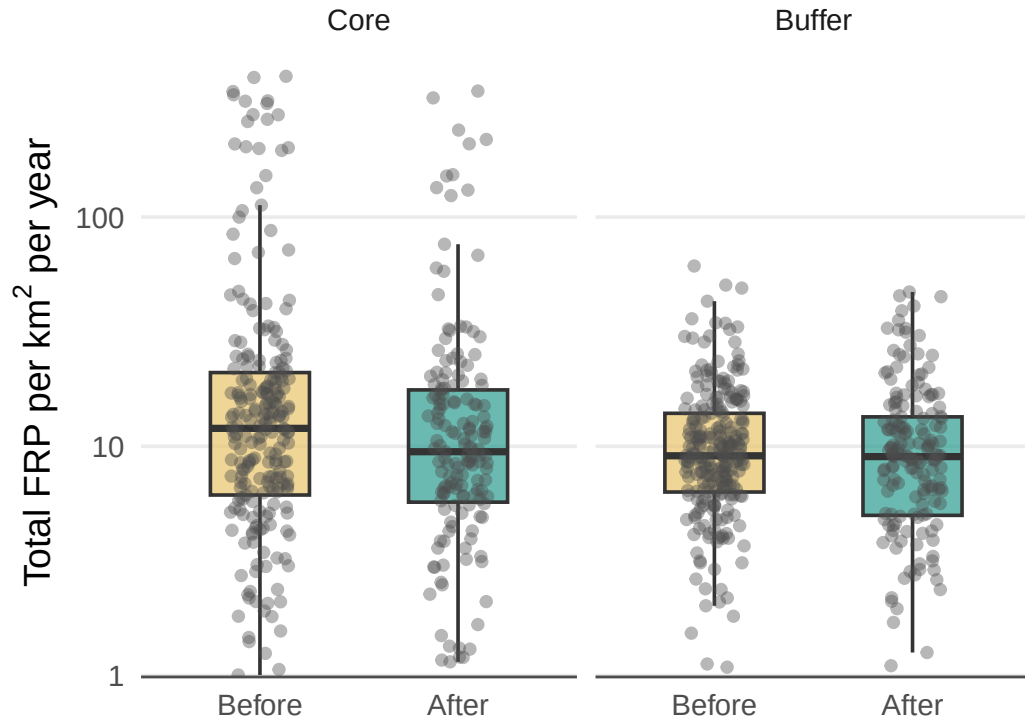


Figure 3: Annual total fire radiative power (FRP) per km² per CFMG before and after establishment, by zone. Total FRP summed across all detections within each CFMG-year-zone and divided by zone area. Each point represents one CFMG-year. Boxes show median and interquartile range. Buffer zone results provide a leakage check.

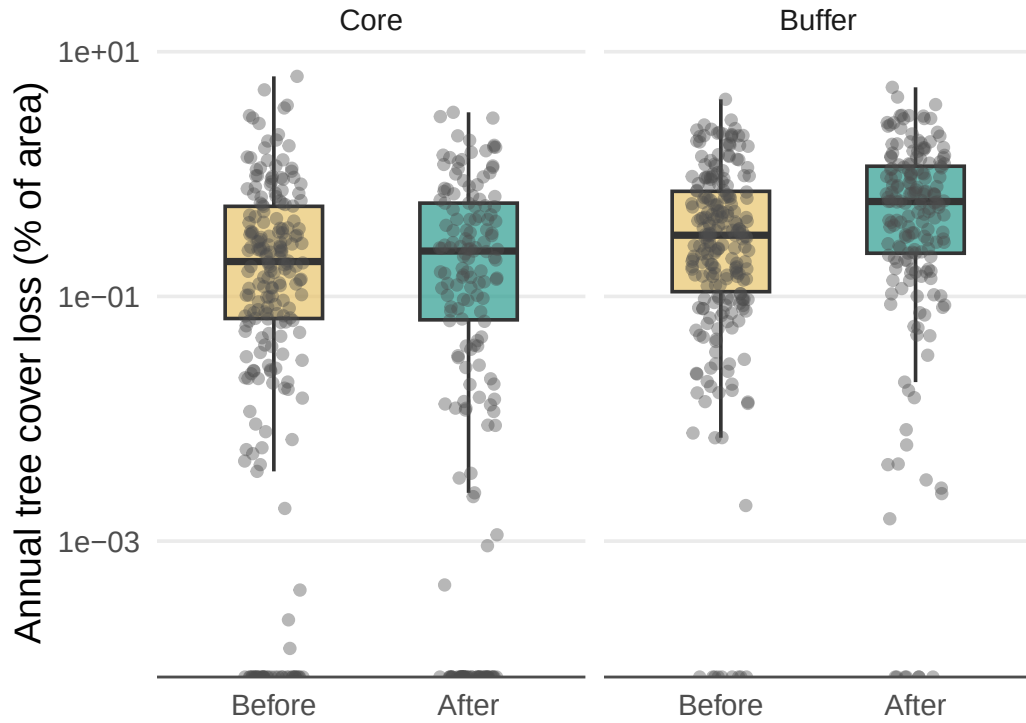


Figure 4: Annual tree cover loss (% of area) in CF cores and 1 km buffer zones before and after CF establishment. Loss derived from Hansen GFC 2024 lossyear data (2014–2024). Each point represents one CFMG-year. Boxes show median and interquartile range.

Table 4: Linear mixed model results for log-transformed annual tree cover loss rate in CF cores and buffer zones before and after establishment.

Model results for annual tree cover loss rate (log-transformed) — core vs buffer, before vs after

Linear mixed model with random intercept for CFMG. Response = $\log(\text{annual loss \%} + 1)$. Fixed effects: period (ref = Before), zone (ref = Core), their interaction, and governance index. A significant After $\times$ Buffer interaction indicates leakage. CFMG variance = 0.065; residual variance = 0.063. Marginal $R^2 = 0.086$; conditional $R^2 = 0.55$. $n = 803$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before, Core)	0.323	0.106	0.0044
After	-0.010	0.026	0.7087
Buffer	0.103	0.024	0.0000
Governance index	-0.015	0.020	0.4737
After $\times$ Buffer	0.170	0.036	0.0000

Deforestation

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The rate of tree cover loss, as measured using the Hansen Global Forest Watch data set, did not change within the CF zones following the establishment of community forestry, but increased in the 1 km buffer zones. The Governance Index did not have any significant effect.

Above ground biomass

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Above ground biomass, as measured using radar backscatter, increased in both CF core areas and their buffer zones following establishment of community forestry. There was no significant effect of Governance Index.

Discussion

Across all three fire metrics — detection density, burned area fraction, and fire radiative power — there were significant reductions in CF cores following establishment. These results provide robust, satellite-based confirmation that CFM has led to meaningful improvements in fire management. Importantly, in none of the fire models was the After $\times$ Buffer interaction significant,

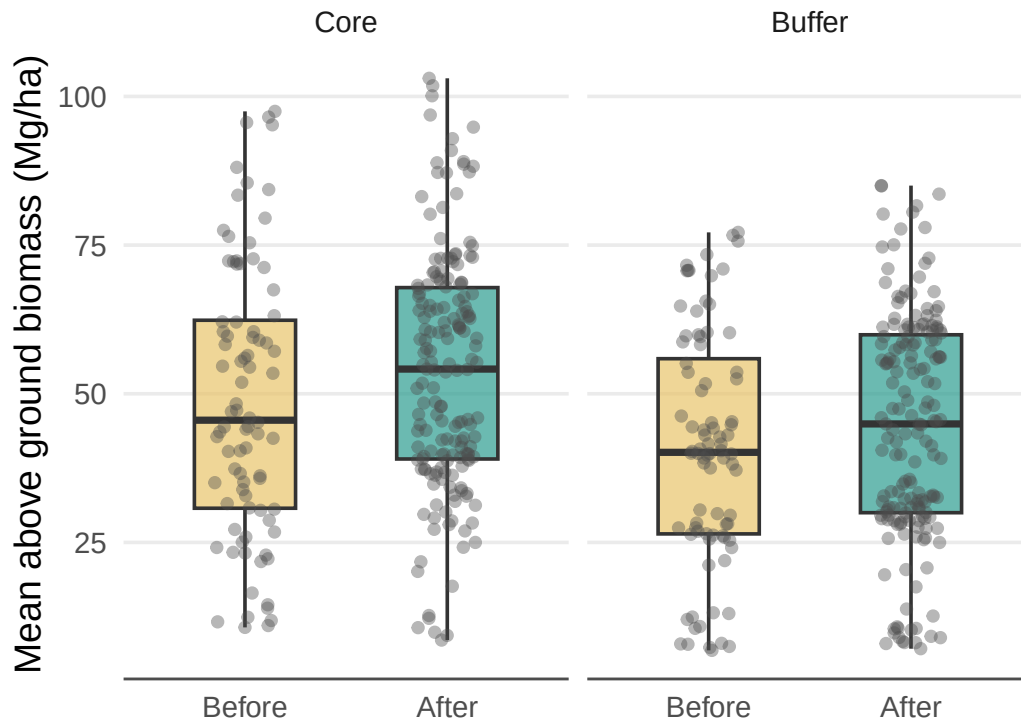


Figure 5: Mean above ground biomass (Mg/ha) per CF before and after establishment, for CF cores and 1 km buffer zones. Values are annual means from 2018–2024; core values are area-weighted across polygons. Each point represents one CF-year. Boxes show median and interquartile range.

Table 5: Linear mixed model results for mean above ground biomass in CF cores and buffer zones before and after establishment.

Model results for mean above ground biomass — core vs buffer, before vs after establishment

Linear mixed model with random intercept for CF. Fixed effects: period (ref = Before), zone (ref = Core), their interaction, and governance index. A significant After $\times$ Buffer interaction indicates leakage. CF variance = 363.972; residual variance = 59.557. Marginal $R^2 = 0.052$; conditional $R^2 = 0.867$. $n = 518$.

Factor	Estimate	SE	<i>P</i> value
Intercept (Before, Core)	49.880	7.815	0.0000
After	4.480	1.087	0.0000
Buffer	-7.974	1.191	0.0000
Governance index	-0.181	1.482	0.9035
After $\times$ Buffer	-1.302	1.449	0.3692

indicating that there was no measurable displacement of burning into the surrounding buffer zones. This absence of leakage suggests that community fire management extends beyond the CF boundary, and is consistent with the SFM chapter finding that most communities also reported a reduction in fire outside their CF.

According to the Hansen data, tree cover loss rates in CF cores did not decrease significantly following establishment. This is in direct contrast to the community perceptions documented in the preceding chapter, where a clear majority of respondents reported that deforestation had decreased within their CF, and also with the results obtained for biomass which showed that biomass increased in both CF core areas and buffer zones. Several explanations for this discrepancy are possible. First, at small spatial scales tree cover data are often not reliable for miombo woodlands, particularly in situations where tree cover is sparse, and can vary seasonally simply according to the relative strength of the grass versus tree canopy signal. Second, the median rates of tree cover loss within the CF core area varied from 0.14% before establishment to 0.13% afterwards. Thus, the deforestation rate was already quite low before establishment and the change is in the desirable direction, even if it was small and not significant. Meanwhile, communities may perceive improvements through indirect indicators — such as reduced charcoal production, fewer incursions, or recovering understorey — that are not captured by the Hansen Global Forest Canopy loss metric. The significant increase in deforestation in the buffer zones from a median of 0.26% before to 0.49% afterwards is a concern. However, as mentioned above canopy cover is not always reliably measured, especially where tree cover is sparse. Thus, this result should be interpreted with caution and tracking tree cover change over a longer period is recommended.

For dry seasonal forests, we anticipate that estimations of biomass change are likely to be a more reliable indicator than tree cover change. However, it is also important to note that they measure different attributes. Biomass monitoring is able to detect small changes caused, for example, through the harvesting of individual trees, but tree growth across the entire zone may cancel out small losses through cutting of individual trees. Nevertheless, above-ground biomass increased significantly in CF cores and buffer zones after establishment. The biomass gain is consistent with — and may be largely attributable to — the reductions in fire disturbance documented above. Fire both kills trees directly and suppresses recruitment of woody species, so even modest reductions in annual burning can lead to measurable biomass accumulation.

The Governance Index was not a significant predictor of any outcome, in contrast to the SFM chapter where governance quality strongly predicted community perceptions of reduced fire frequency and deforestation. It is also possible that basic fire management is one of the first activities initiated by newly established CFMGs, and that even groups with weaker governance have put fire patrols in place.

Taken together, the remote sensing evidence presents a nuanced picture of CFM effectiveness in Zambia. CFM has demonstrably reduced fire disturbance and, in consequence, has produced measurable gains in above-ground biomass — without evidence of displacement into surrounding buffer areas. However, there has not been any measurable decrease in deforestation — as detected through the Global Forest Canopy dataset — and there is an indication that deforestation may have increased in buffer areas. While the deforestation result should be investigated further, it may suggest that the economic drivers of deforestation remain largely unaddressed by current CFM practices.

- Giglio, L., W. Schroeder, and C. O. Justice. 2016. “The Collection 6 MODIS Active Fire Detection Algorithm and Fire Characterization Study.” *Remote Sensing of Environment* 178: 31–41. <https://doi.org/10.1016/j.rse.2016.02.054>.
- Hansen, M. C., P. V. Potapov, R. Moore, M. Hancher, S. A. Turubanova, A. Tyukavina, D. Thau, et al. 2013. “High-Resolution Global Maps of 21st-Century Forest Cover Change.” *Science* 342 (6160): 850–53. <https://doi.org/10.1126/science.1244693>.
- Harrison, S. B., J. L. Godlee, D. T. Milodowski, et al. in prep. “Carbon Accumulation in Tropical Dry Forests and Savannas Despite Widespread Land Cover Change, Revealed by Radar Remote Sensing.”